

Wafer Map Defect Classification with Machine Learning

Najani Johnson

University at Albany - College of Nanoscale,
Science and Engineering

Meghan Herbert

University at Albany - College of Nanoscale,
Science and Engineering



UNIVERSITY
AT ALBANY
STATE UNIVERSITY OF NEW YORK



U A L B A N Y SHOWCASE

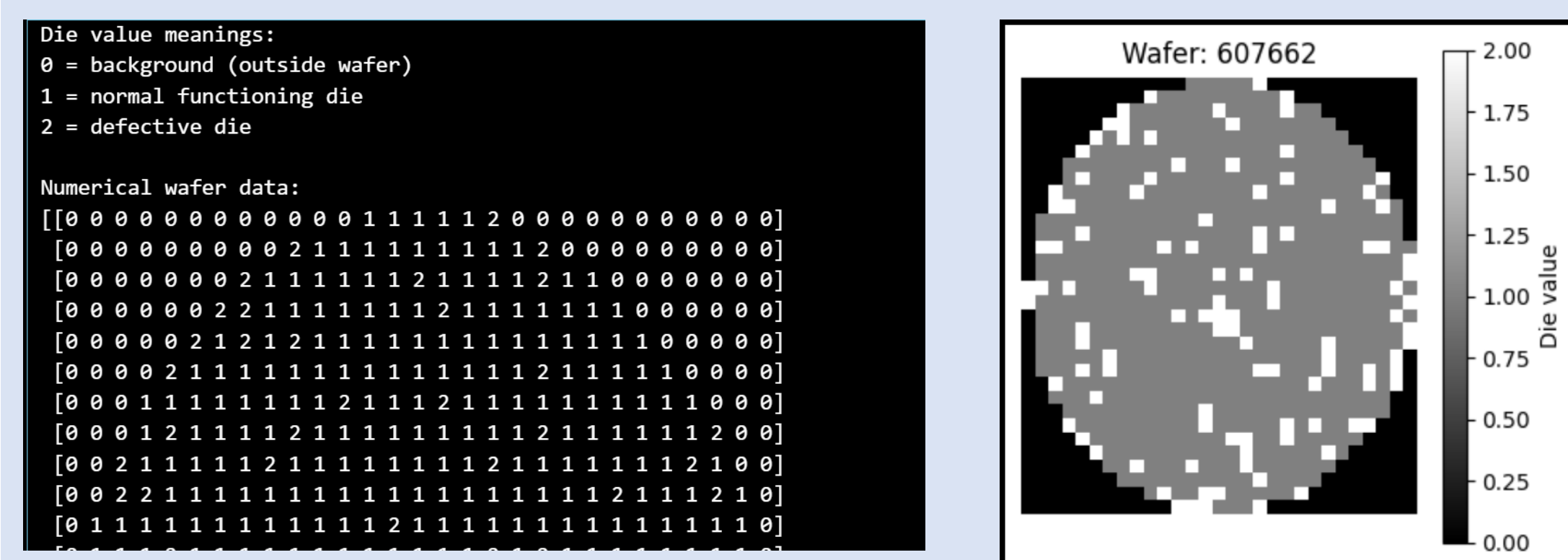
Background

- Semiconductor wafers contain thousands of dies
- These defects reduce total manufacturing yield.
- Manual inspection of wafers is slow and expensive, therefore using machine learning we can approximate

Dataset

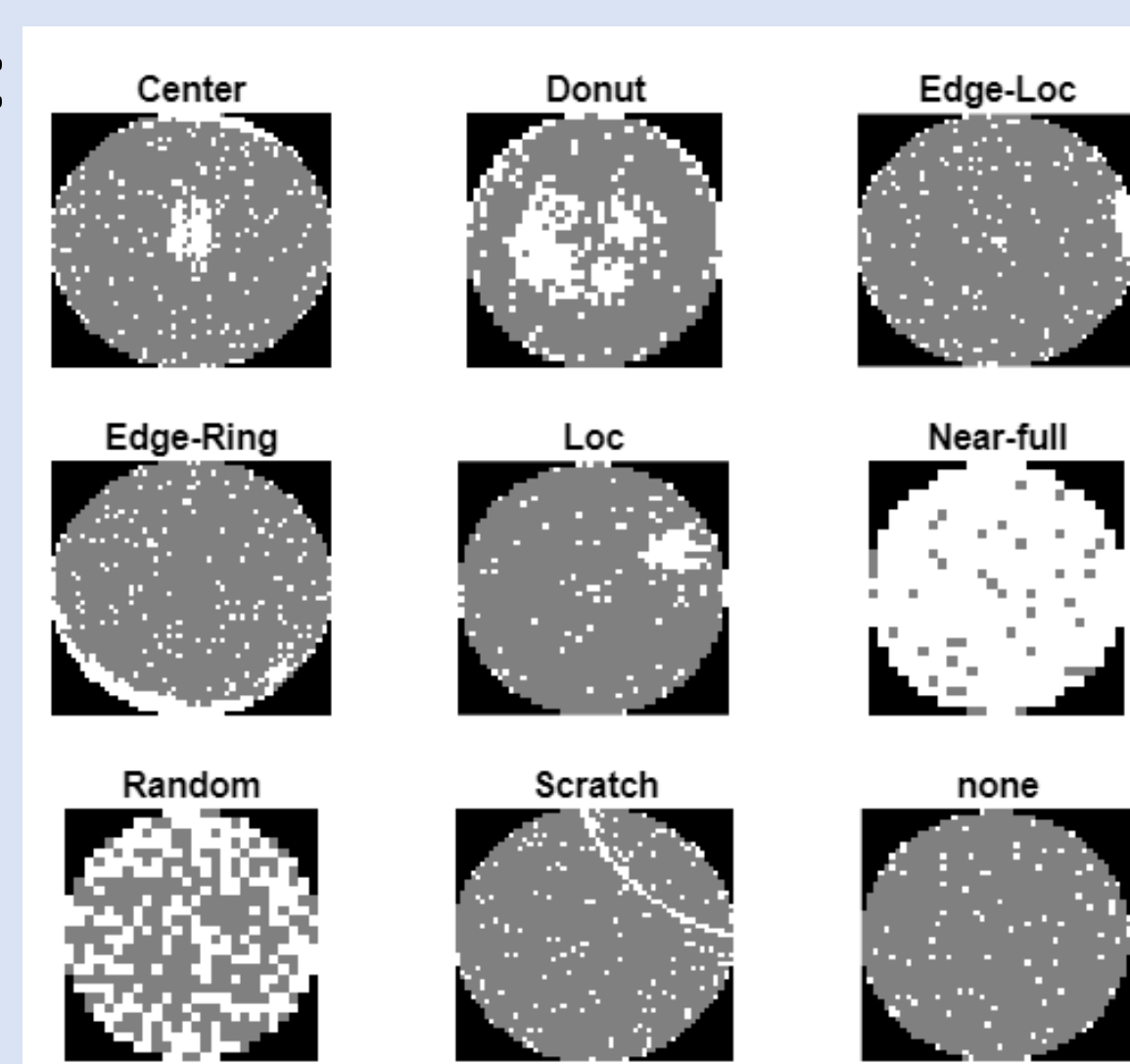
- For this project, we decided to go with the WM-811K Wafer Map dataset, which contains many real wafer maps collected from semiconductor manufacturing.
- This data comes pre-classified, alongside with test and training datasets.

Wafer Maps



Wafer Data

- The actual wafer data is stored in a PKL file which can be decompressed using Python to access the individual 811,000 data sets
- Each of these has a value for the individual dies failing or passing, alongside the actual dimension to re construct the original wafer.
- In terms of actual classifications, we have the following types:



Incorporating Random Forest Model

Purpose

- Used as a baseline model for comparison
- Provides a reference to evaluate CNN performance

Method

- Wafer maps flattened into 1D feature vectors
- Trained using Random Forest (100 trees)

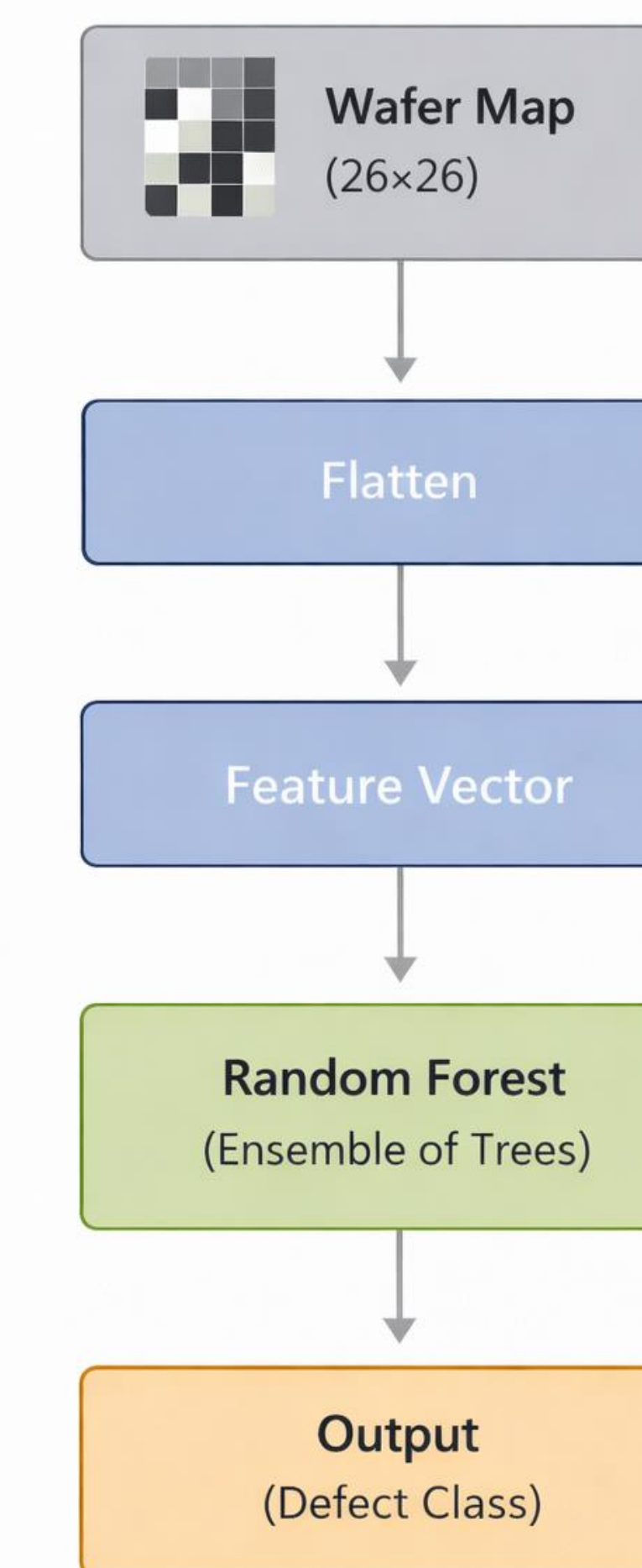
No spatial feature learning

Evaluation

- Metrics: Accuracy, Precision, Recall, F1-score
- Helps identify misclassified defect types

Limitation

- Cannot capture spatial patterns in wafer images
- Motivates use of CNNs for improved performance



Incorporating CNN Model

Purpose

- Capture spatial patterns in wafer maps
- Improve performance over traditional ML models

Method

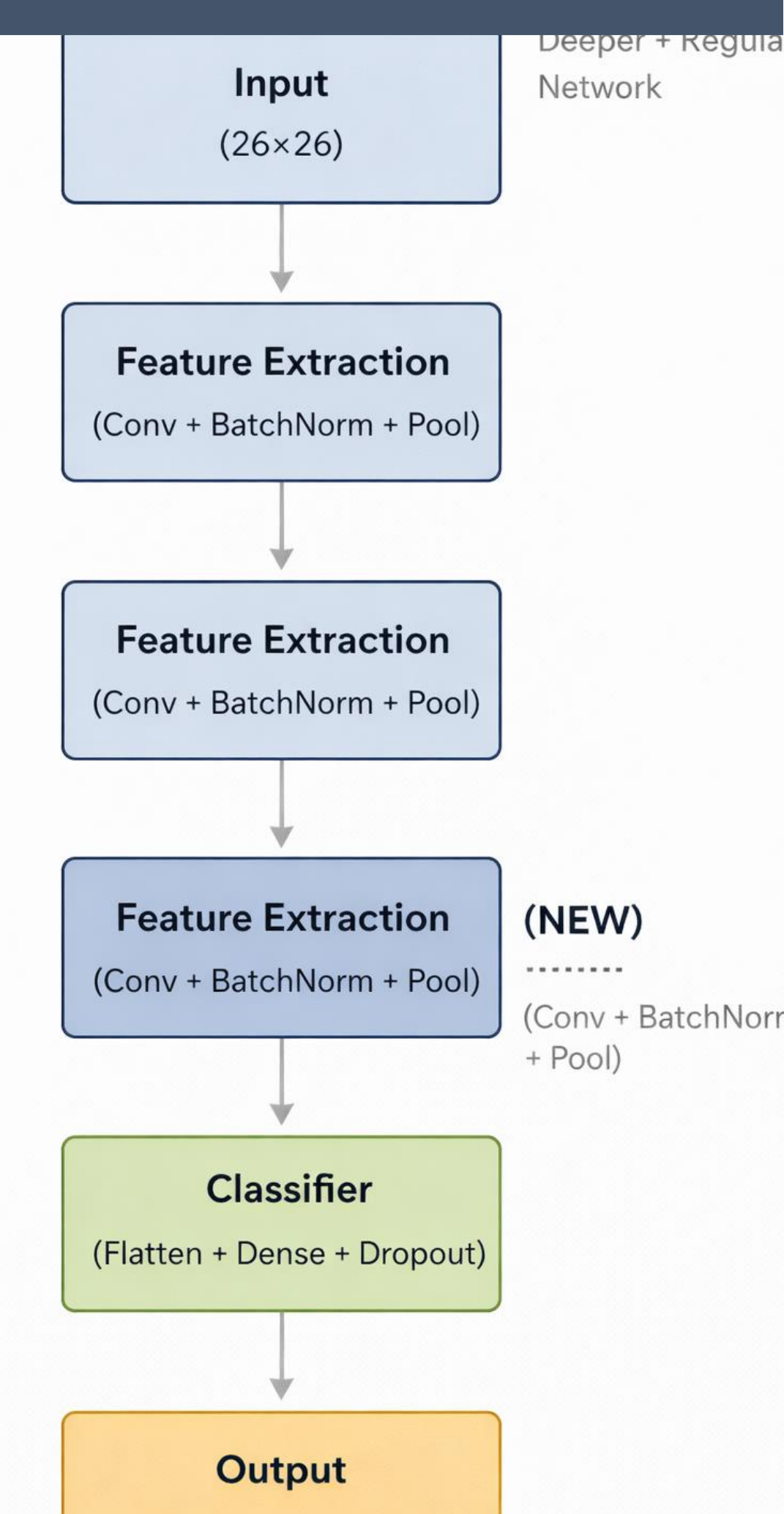
- Input: 26×26 wafer images
- Deep CNN architecture (see diagram)

Evaluation

- Metrics: Accuracy, Precision, Recall, F1-score
- Compared against Random Forest baseline
- Visualized using confusion matrix & training curves

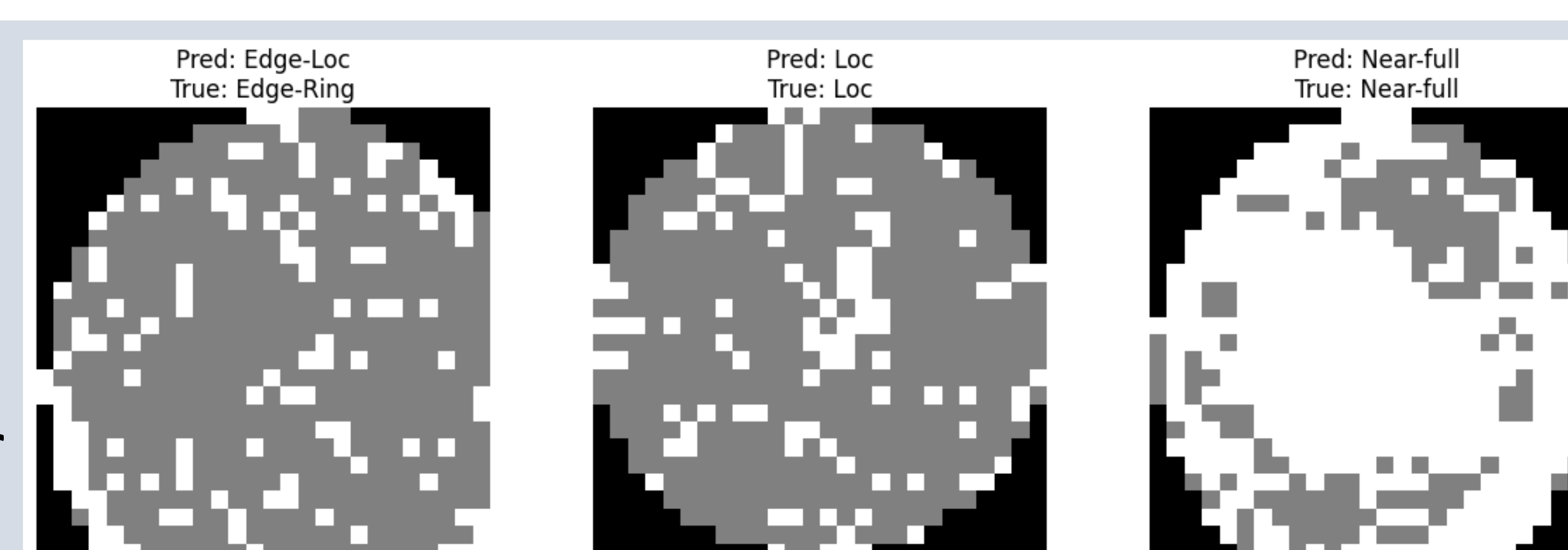
Improvement Over Baseline

- Learns spatial relationships in images
- Better detection of defect patterns
- More suitable for image-based classification

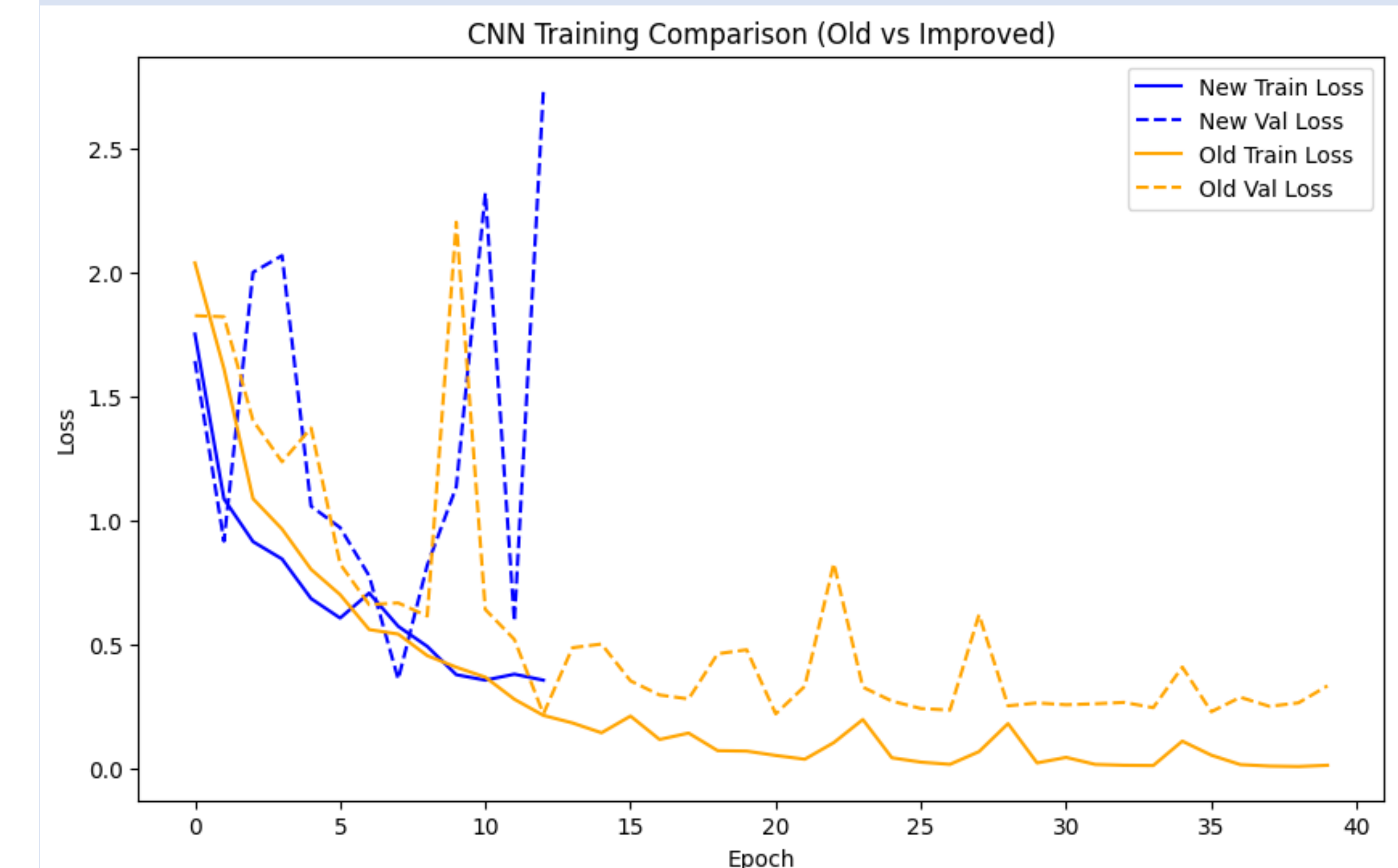


Sample Results:

2 out of 3 samples
correctly classified;
misclassification occurs
between visually similar
defect patterns.



Key Takeaways



- Improved CNN stops early preventing further overfitting, while the old model continues training and shows increasing validation loss
- Old model achieves lower training loss but worse generalization which directly contrasts the improved model

	Model	Accuracy	F1 Score
0	Random Forest	0.9592	0.9479
1	CNN (Old)	0.9663	0.9663
2	CNN (Improved)	0.9646	0.9614

- CNN models outperform Random Forest, with the improved CNN achieving the highest accuracy while maintaining strong overall performance

References

- [1] Wu, M. J., et al., “WM-811K Wafer Map Dataset,” Kaggle.
- [2] Breiman, L., “Random Forests,” Machine Learning, 45(1), 5–32, 2001.
- [3] Goodfellow, I., Bengio, Y., and Courville, A., Deep Learning. MIT Press, 2016.
- [4] LeCun, Y., Bengio, Y., and Hinton, G., “Deep learning,” Nature, 521, 436–444, 2015